Chengtai Cao

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Publication

Conference paper:

1. **Overcoming catastrophic forgetting in graph neural networks with experience replay. (Corresponding Author)** **AAAI 2021 (CCF-A)**

   *Fan Zhou, **Chengtai Cao****

   - We conduct an empirical investigation of the continual node classification task, demonstrating that existing GNNs are in the dilemma of catastrophic forgetting when learning a stream of tasks in succession.
   - To address the catastrophic forgetting issue, we develop a generic experience replay based framework that can be easily combined with any popular GNNs model. Apart from two intuitive experience selection schemes, we propose a novel strategy based on influence function.
   - We conduct extensive experimental evaluations using three benchmarks to demonstrate our framework's superiority over several state-of-the-art GNNs.

2. **Meta-GNN: on few-shot node classification in graph meta-learning. (First author of the students)** **CIKM 2019 (CCF-B)**

   *Fan Zhou, **Chengtai Cao**, Kunpeng Zhang, Goce Trajcevski, Ting Zhong, Ji Geng*

   - We formulate a new graph few-shot learning paradigm for node classification. It is different from previous work in that we aim to classify nodes from new classes with only few samples each.
   - We propose a generic framework for tackling few-shot node classification, which can be easily combined with any popular GNNs models and opens up a new perspective of graph structure data analysis.
   - We demonstrate the superiority of our method over several state-of-the-art GNNs models on three benchmarks.

3. **Fast network alignment via graph meta-learning. (First author of the students)** **INFOCOM 2020 (CCF-A)**

   *Fan Zhou, **Chengtai Cao**, Goce Trajcevski, Kunpeng Zhang, Ting Zhong, Ji Geng*

   - By recasting the NA as an OSL task, we provide an initial attempt to tackle this problem with meta-learning.
   - Through learning the generalization among mini-NA tasks, Meta-NA enables generalizing the knowledge from linked anchor nodes to unlinked pairs, while easily adaptable to scenarios of aligning multiple networks.
   - Due to the ability of learning fast adaption of mini-NA tasks, Meta-NA is more efficient than existing graph learning based NA approaches.

4. **Trajectory-user linking via graph neural network.** **ICC 2021 (CCF-C)**

   *Fan Zhou, Shupei Chen, Jinwu, **Chengtai Cao**, Shengming Zhang*

Publication

Journal Paper：

Learning meta-knowledge for few-shot image emotion recognition
(First author of the students). **ESWA 2021 (Q2, IF: 6.954)**
Fan, Zhou, ***Chengtai Cao****, Ting Zhong, Ji Geng*

- We formulate a new few-shot learning paradigm for image emotion classification and propose a generic framework called Meta-IEC.
- we conduct extensive experiments on three commonly used image emotion datasets, demonstrating our method's superiority over several state-of-the-art baselines

Education

University of Electronic Science and Technology of China 2018.09 – 2021.06
Master of Software Engineering (GPA 3.79/4)
University of Electronic Science and Technology of China 2014.09 – 2018.06
Bachelor of Software Engineering (GPA 3.06/4)

Honors and Awards

National Scholarship for two consecutive years (Top 9/1000) 2019, 2020
Excellent Master Thesis (UESTC) 2021
People First Prize Scholarship for two consecutive years (Top 2/700) 2019, 2020
Excellent practice Team (Team Leader) 2016
Merit Student (Sichuan Province) 2014

Work Experience

Baidu International Technology Co., Ltd. (Shenzhen) 2021.07 – Present Strategy Algorithm Engineer
I am working for optimizing the rank model in the recommendation system of Ruliu iteratively and the rank model is Wide&Deep model now.
(Main tools: Python, Paddle, and Pytorch)

SAP China Co., Ltd. (Chengdu) 2018.02 - 2018.06 Font-end Developer and QA
I exploited React.js, Redux, and Ant Design Pro to build a CRM APP for Qirui.
(Main tools: HTML, CSS, Javascript, Reat.js, Redux, and Ant Design Pro)